

How TrueCrypt works

TrueCrypt is a great open source tool for this purpose. It supports Windows and Mac platforms, and also supports almost all Linux distros. Typically, on Ubuntu and CentOS systems running in a business-critical data centre, TrueCrypt is widely used to encrypt disk volumes, to keep data safe from physical theft by prying attackers. It works on the principle of 'on the fly' encryption, whereby the encryption and decryption operations are performed once the authentication key is provided, and the process takes place transparently to the user. To achieve this, the utility installs kernel-level drivers (Linux) or device drivers (Windows) by hooking up with the disk management modules of the OS, thus acting as an agent between disk read/writes and the application layer. TrueCrypt can also create a single encrypted hidden volume inside an existing file-system volume, for elevated security. Since it can handle an entire mounted volume, it can be easily used to encrypt data on a USB pen drive too. As for the crypto processing, the utility does it so fast and seamlessly, that the user doesn't even know about it. Usually, such on-the-fly utilities demand heavy memory and CPU usage, but that's not the case with TrueCrypt. It achieves this by making use of all CPUs and cores available on the system. It also uses a technique called pipelining, whereby a portion of the disk is read and decrypted in memory, acting as a buffer, in anticipation that the application would want to fetch that data piece.

TrueCrypt requires a user-entered password to encrypt the volume, and also allows the selection of an algorithm for encryption. The following table lists the algorithms supported and the typical usage of each.

Encryption Algorithm	Hashing Algorithm
AES (256bit key)	RIPEMD (160bit key)
Serpent (256bit key)	SHA-512 (512bit key)
Twofish (256bit key)	Whirlpool (512bit key)

Besides the keys, on Windows machines TrueCrypt allows the user to generate a random key by moving the mouse randomly. This key is fed to the encryption algorithm to further enhance its strength, thus introducing tighter data encryption security. It is important to note that there is no bypass or backdoor for the utility if the pass-phrase is forgotten. Due to the complex algorithms in use, it can practically take millions of years to crack an encrypted volume in such situations. TrueCrypt can also accept PKCS #11 protocol-based devices such as smart cards or secure-tokens. This makes it a must-have utility for volumes carrying seriously critical corporate data.

Installation

This utility can be downloaded from sourceforge.net/projects/truecrypt/, or the latest available version from <http://www.truecrypt.org>. While installing on GUI-based Ubuntu or a similar distro, just unpack and double-click the binary archive to get it installed under *Applications > Accessories > TrueCrypt*. Figure 1 shows a screen shot of TrueCrypt being installed on an Ubuntu distro. Upon running the utility, it gives you a choice to either create an encrypted container, or a protected volume within a



Figure 1: A screenshot of TrueCrypt being installed on an Ubuntu distro



Figure 2: A sample installation screen on Windows

partition or disk drive mount. The next step is to select encryption and hash algorithms, and the pass-phrases. Figure 2 shows a sample installation screen on Windows. Depending on the system's hardware configuration, it may take a while for TrueCrypt to process the data. This can be significantly long for a high volume of data.

IT administrators are advised to try some hands-on work and gain expertise on the utility in a lab set-up, prior to deploying it on their production infrastructure. This is because there is no mercy or trick to retrieve locked data if the pass-phrase is forgotten. The drive or mounted volume under TrueCrypt will be totally inaccessible under such situations. This also calls for tighter security mechanisms to store the key in a safe location. Since it performs encryption-decryption on the fly, disk fragmentation occurs quite often; hence, for better system performance, administrators are also advised to defrag the volumes periodically. TrueCrypt is installed on business-critical production systems including Web servers, file servers, email servers, etc, which makes it a standard utility for IT infrastructures where business data security is paramount. **END**

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